

Domain Wall Memories with Reconfigurable Pinning by Magnetic Topological Insulator

Arifa Hoque, Sanjukta Bhanja
University of South Florida, Tampa, FL 33612, USA

Domain Wall Memories (DWMs) require periodic pinning sites to precisely control domain-wall (DW) motion. Conventional lithographic notches and geometrical constrictions provide pinning but introduce fabrication variability and fixed pinning strength. This work investigates the use of magnetic topological insulator (MTI) nanobars as electrically reconfigurable pinning sites for ferromagnetic (FM) nanowires. Direct exchange coupling at the FM–MTI interface generates a local pinning potential, while current-induced spin accumulation in the MTI surface states provides a mechanism for controlled depinning. Micromagnetic simulations verify MTI-assisted DW pinning and demonstrate the feasibility of tunable pinning at the pinning sites.

Index Terms—Domain wall memory, Spintronics, Emerging memories, Reconfigurable Memories

I. INTRODUCTION

In conventional DWMs, lithographic notches are periodically introduced along a ferromagnetic nanowire to create predefined pinning sites that hold domain walls between magnetic domains, ensuring reliable device operation [1]. Although effective, these approaches create fixed pinning potentials, restricting DWMs to conventional storage or Boolean in-memory computing [2]. In contrast, DWMs with reconfigurable pinning sites offer variable storage capacity and support near-analog in-memory computation, which can benefit AI/ML applications. Thus, a reconfigurable pinning approach is desirable, where the pinning potential can be electrically controlled without topographical deformation in the nanowire.

Magnetically doped topological insulators (MTIs) are attractive for this purpose because their spin-polarized surface states can exchange-couple with an adjacent ferromagnet, enabling local control of DW pinning. This work proposes MTI nanobars placed on top of an FM nanowire as controllable pinning elements, as shown in Fig. 1a. At each MTI–FM cross-point, the exchange interaction creates a local pinning potential, while current-induced spin accumulation in the MTI surface enables localized torque-assisted depinning. The proposed structure therefore provides a notch-free, electrically tunable pinning mechanism for reconfigurable DWM architectures.

II. MTI ASSISTED DOMAIN-WALL PINNING

In a three-dimensional undoped topological insulator (TI), the bulk is insulating, whereas the surface hosts highly conductive gapless electronic states. These gapless states are protected by time-reversal symmetry (TRS), making them robust against non-magnetic perturbations [4]. Introducing magnetism through magnetic doping breaks TRS and opens a surface-state gap. The resulting magnetic topological insulator (MTI) can exchange-couple with an adjacent ferromagnet, enabling local modulation of magnetization and DW pinning. Since charge transport in a three-dimensional topological insulator occurs mainly through surface states, the physics is described by the low-energy surface Hamiltonian near the Dirac (Γ) point. The gapless surface Hamiltonian at the MTI–FM interface, neglecting any tunneling through the MTI is

$$H_0 = \hbar v_F (k_y \sigma_x - k_x \sigma_y), \quad (1)$$

where v_F is the Fermi velocity, $\vec{k} = (k_x, k_y)$ is the in-plane wave vector, and σ_x , σ_y , and σ_z are the Pauli spin matrices.

This Hamiltonian represents helical spin–momentum-locked surface states, where electron spin is locked perpendicular to momentum. The eigenvalues are $E_{\pm}(\vec{k}) = \pm \hbar v_F \sqrt{k_x^2 + k_y^2}$, corresponding to a gapless Dirac cone at $\vec{k} = 0$ (Fig. 1a).

When perpendicular magnetic dopants are introduced, time-reversal symmetry is broken through exchange interaction between the surface-state spin and dopant magnetic moments. For PMA, the MTI magnetization is along the out-of-plane direction, $M_z \hat{z}$, introducing an exchange-induced mass term, $m_z \sigma_z$, where m_z is proportional to the out-of-plane magnetic order. Thus, the PMA-MTI surface Hamiltonian becomes

$$H_{MTI} = \hbar v_F (k_y \sigma_x - k_x \sigma_y) + m_z \sigma_z. \quad (2)$$

The corresponding energy dispersion is

$$E_{\pm}(\vec{k}) = \pm \sqrt{(\hbar v_F)^2 (k_x^2 + k_y^2) + m_z^2}. \quad (3)$$

Therefore, the originally gapless Dirac cone becomes gapped, with an exchange gap of $\Delta = 2|m_z|$ at $\vec{k} = 0$. This exchange-induced gap provides intrinsic magnetic order in the PMA-MTI and enables deterministic exchange coupling with the adjacent PMA ferromagnetic nanowire.

At the FM–MTI interface, the PMA-MTI magnetization couples to the local magnetization of the ferromagnetic nanowire. The interfacial exchange energy is

$$E_{int} = -J_{int} \vec{m}_{FM} \cdot \vec{m}_{MTI}, \quad (4)$$

where J_{int} is the interfacial exchange coefficient, $\vec{m}_{FM}$ and $\vec{m}_{MTI}$ are the unit magnetization of the NW and the MTI. Since $\vec{m}_{MTI}$ is fixed along $\hat{z}$, the interfacial energy depends on the local out-of-plane magnetization component of the nanowire, which has domains magnetized along $+\hat{z}$ or $-\hat{z}$, and the domain walls containing a rotating magnetization profile. When the wall reaches the MTI-covered region, the exchange energy becomes spatially nonuniform, creating an additional energy barrier at the FM–MTI overlap. This barrier forms a stable pinning site without requiring a lithographic notch. The total nanowire energy can be written as

$$E_{tot} = E_{ex}^{FM} + E_{ani}^{FM} + E_{demag}^{FM} + E_{int}, \quad (5)$$

where the first three terms represent the exchange, anisotropy, and demagnetization energies of the FM nanowire. The E_{int} term exists only at the MTI–FM cross-point and modifies

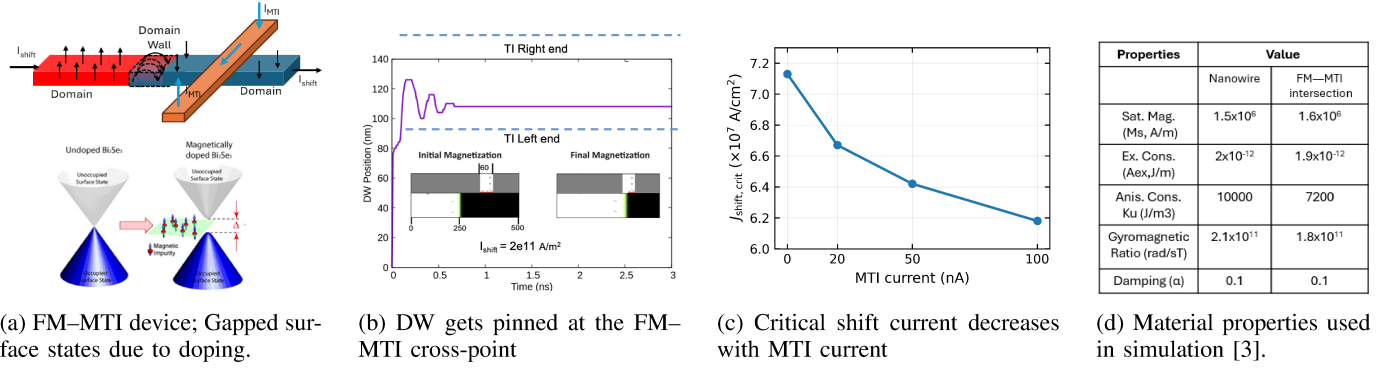


Fig. 1: Demonstration of domain-wall pinning with variable strength at the FM-MTI interface due to direct exchange interaction.

the local DW energy. As a result, the wall experiences a localized pinning potential whose strength depends on the MTI magnetization, exchange coefficient, and overlap area.

In addition to static pinning, the PMA-MTI can assist depinning through current-induced spin accumulation. When an in-plane current flows through the MTI surface, spin-momentum locking generates a nonequilibrium spin density. This spin density interacts with the domain wall through exchange coupling and produces a localized spin torque at the MTI-FM interface. The field-like torque is proportional to the exchange coupling and the non-equilibrium spin density, and can be written as

$$\vec{T}_{FL} = \Delta_{ex}(\vec{m}_{FM} \times \vec{S}_d), \quad (6)$$

where $\vec{S}_d$ is the diffused spin density in the ferromagnet. The spatial variation of spin current in the ferromagnet produces a Slonczewski-like torque. For spin diffusion along the z -direction, the torque is expressed as [5]

$$T_{STT,\parallel} + iT_{STT,\perp} = \frac{pJ_s L^2}{d} \left(\frac{1}{\lambda_\phi^2} - \frac{i}{\lambda_j^2} \right) \frac{\cosh(\frac{d}{L}) - 1}{\sinh(\frac{d}{L})} \quad (7)$$

L in the above equation is a function of the spin relaxation length (λ_{sf}), spin precession length (λ_j), and spin decoherence length (λ_ϕ), and is defined as $\frac{1}{L^2} = \frac{1}{\lambda_{sf}^2} + \frac{1}{\lambda_\phi^2} - \frac{i}{\lambda_j^2}$. The corresponding effective torque field is $\vec{H}_{STT} = T_{STT}(\vec{m} \times \hat{\sigma})$, where $\hat{\sigma} = \hat{j} \times \hat{k}$. Since this torque is localized at the MTI-FM overlap, it does not independently drive steady DW motion along the nanowire. Instead, it reduces the local pinning barrier during shift operation, enabling controlled depinning when a shift current is applied through the nanowire.

III. SIMULATION RESULTS AND DISCUSSION

Micromagnetic simulations were performed using MuMax3 [6] with the material properties listed in the table in Fig. 1d. The simulated device consists of a 500 nm long, 100 nm wide, and 2 nm thick FM nanowire oriented along the x -direction. A 60 nm-wide MTI nanobar was positioned on the right half of the wire, with 100 nm center-to-center spacing between FM and MTI. The PMA nanowire used $\text{Co}_{40}\text{Fe}_{40}\text{B}_{20}$. MTI nanobars were modeled using Cr-doped $(\text{Bi}_{0.5}\text{Sb}_{0.5})_2\text{Te}_3$. The simulation cell size was $2 \times 2 \times 2 \text{ nm}^3$. The MTI magnetization was initialized along $(0, 0, 1)$, and a DW was initially placed at the center of the nanowire. To verify MTI-induced pinning, a current pulse of $2 \times 10^7 \text{ A/cm}^2$ with 3 ns

duration was applied along the $-x$ direction. The DW starts propagating under the applied current. With the MTI nanobar present with no current flowing through it, the DW moves from its initial position and becomes pinned at the MTI-FM intersection after approximately 100 nm displacement (Fig. 1b). This confirms that the exchange interaction at the MTI-FM cross-point creates a local energy barrier that holds the DW during shift operation.

For DW depinning, the MTI-induced torque is incorporated as a localized effective field at the MTI-FM intersection. The current through the MTI layer is varied as 0, 20, 50, and 100 nA, and the corresponding critical shift current is calculated. Using spin efficiency $p = 1$, $\lambda_{sf} = 5 \text{ nm}$, $\lambda_\phi = 1 \text{ nm}$, and $\lambda_j = 1 \text{ nm}$ [7], the in-plane and out-of-plane torques are calculated from Eq. 7 for each MTI current and incorporated into the simulation as localized fields. With no MTI current, the torque on the DW is zero, resulting in the highest pinning potential. As the MTI current increases, the induced torque assists DW depinning from the MTI-FM cross-point. Although this torque alone is insufficient to drive steady DW motion along the nanowire, it reduces the shift current required for depinning. Fig. 1c shows that the critical shift current decreases with increasing MTI current. Therefore, controlling the MTI current enables modulation of the pinning potential and makes the DWM reconfigurable.

REFERENCES

- [1] S.-H. Huang and C.-H. Lai, "Domain-wall depinning by controlling its configuration at notch," *Applied Physics Letters*, vol. 95, no. 3, 2009.
- [2] X. Zhou, Z. Huang, W. Zhang, Y. Yin, P. Dürrenfeld, S. Dong, and Y. Zhai, "Current-induced multiple domain wall motion modulated by magnetic pinning in zigzag shaped nanowires," *AIP Advances*, vol. 7, no. 5, 2017.
- [3] A. K. Reza, "Modeling and simulation of topological insulators, topological semi-metals and ferrimagnets for time and energy efficient switching of magnetic tunnel junction," Ph.D. dissertation, Purdue University, 2019.
- [4] C.-X. Liu, X.-L. Qi, H. Zhang, X. Dai, Z. Fang, and S.-C. Zhang, "Model hamiltonian for topological insulators," *Physical Review B*, vol. 82, no. 4, p. 045122, 2010.
- [5] A. Manchon, R. Matsumoto, H. Jaffres, and J. Grollier, "Spin transfer torque with spin diffusion in magnetic tunnel junctions," *Physical Review B*, vol. 86, no. 6, p. 060404, 2012.
- [6] A. Vansteenkiste, J. Leliaert, M. Dvornik, M. Helsen, F. Garcia-Sanchez, and B. Van Waeyenberge, "The design and verification of mumax3," *AIP advances*, vol. 4, no. 10, p. 107133, 2014.
- [7] A. Mellnik, J. Lee, A. Richardella, J. Grab, P. Mintun, M. H. Fischer, A. Vaezi, A. Manchon, E.-A. Kim, N. Samarth *et al.*, "Spin-transfer torque generated by a topological insulator," *Nature*, vol. 511, no. 7510, pp. 449–451, 2014.